

# Mycelium -- Semantic Architecture Consolidation Notes

## Purpose

This document consolidates all major architectural discussions, proposed improvements, unresolved questions, and future implementation directions discussed regarding:

- Predicate schemas
- Semantic routing
- MultiLensRouter evolution
- TRM integration
- Claim decomposition
- Temporal reasoning
- Confidence propagation
- Ontology stabilization
- DAG reasoning structures
- Contradiction semantics
- Evidence lineage
- Canonical graph formation

The goal is to maintain a single unified reference for future implementation planning.

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## 1. Core Architectural Realization

Mycelium is no longer evolving as merely:

- an expert routing system,
- an LLM orchestration system,
- or a multi-model inference architecture.

It is evolving into:

| A semantic reasoning substrate with epistemic stabilization.

The system is increasingly focused on:

- semantic arbitration,
- contradiction reasoning,

- ontology reconciliation,
  - evidence-grounded cognition,
  - longitudinal semantic stabilization,
  - and uncertainty-aware reasoning.
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## 2. Predicate Schema Architecture

### Original Problem

Different experts or decomposition pipelines may produce semantically equivalent but structurally different predicates.

Example:

```
Orbits(Earth, Sun)
MovesAround(Earth, Sun)
OrbitalMotion(Earth, Sun)
```

Without canonicalization:

- caching fragments,
  - contradiction analysis weakens,
  - graph reuse collapses,
  - evidence retrieval becomes inconsistent.
- 

### Proposed Solution

#### Dynamic Predicate Schemas

Instead of forcing one rigid predicate structure globally:

- allow multiple schema variants,
- then use semantic arbitration to reconcile them.

This responsibility can be delegated to:

- MultiLensRouter
- TRM stabilization layer

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## Canonical Predicate Families

Instead of collapsing everything directly:

```
Causes
IncreasesRisk
AssociatedWith
CorrelatesWith
Predicts
```

into a single predicate,

organize them into:

```
RELATION_FAMILY: CAUSAL
```

with subtype distinctions preserved.

This prevents dangerous semantic collapse.

Example:

```
Causes(Smoking, Cancer)
```

is NOT equivalent to:

```
IncreasesRisk(Smoking, Cancer)
```

One implies deterministic causality.

The other implies probabilistic association.

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## 3. Dynamic Argument Structure

### Proposed Representation

Instead of fixed-arity predicates:

```
Relation(subject, object)
```

use graph-style dynamic structures.

Example:

```
func(  
  entities=[Earth, Sun],  
  relationships={  
    (Earth, Sun): "orbits"  
  }  
)
```

---

## Benefits

This allows:

- dynamic arity,
- graph-native decomposition,
- compositional reasoning,
- semantic flexibility,
- ontology extensibility.

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## Important Realization

NER alone is insufficient.

Future implementations should include:

- Semantic Role Labeling (SRL)
- Dependency Parsing
- Event Extraction
- Relation Extraction
- MultiLensRouter-assisted semantic arbitration

because entities alone do not preserve semantic roles.

Example:

```
John gave Mary a book.
```

Needs:

- giver,
  - receiver,
  - transferred object,
  - transfer relationship.
- 

## 4. Temporal Encoding System

### Initial Problem

Static timestamps are insufficient.

The system must support:

- exact timestamps,
  - intervals,
  - uncertain time,
  - relative temporal relations,
  - historical validity.
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### Proposed Structure

Instead of:

N/A

use structured temporal states.

Example:

```
temporal_state = {  
  "type": "UNKNOWN"  
}
```

Possible future temporal types:

```
"EXACT"  
"INTERVAL "  
"APPROXIMATE"  
"RELATIVE "  
"HISTORICAL_ESTIMATE"  
"UNKNOWN"
```

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## Relative Temporal Reasoning

Example:

```
Smoking preceded diagnosis
```

Should become:

```
BEFORE(smoking, diagnosis)
```

The MultiLensRouter may assist in resolving:

- relative ordering,
  - temporal semantics,
  - uncertainty interpretation.
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## Historical Validity

Claims may change over time.

Example:

```
PlutoIsPlanet
```

True historically,  
false after reclassification.

Temporal validity tracking is required for:

- reasoning cache integrity,
- contradiction analysis,

- ontology evolution.
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## 5. Confidence Encoding

### Initial Decision

Confidence should become part of graph structures.

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### Recommended Structure

Instead of:

```
0.82
```

use structured confidence metadata.

Example:

```
confidence = {  
  "score": 0.82,  
  "type": "epistemic",  
  "source": "external_evidence",  
  "method": "bayesian_aggregation"  
}
```

---

### Future Confidence Types

Eventually confidence should separate into:

```
semantic_confidence  
structural_confidence  
epistemic_confidence  
evidence_confidence  
temporal_confidence
```

because these represent fundamentally different uncertainty dimensions.

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## 6. Claim Decomposition System

### Original Problem

Natural language must become:

- structured claims,
- predicates,
- evidence hooks,
- reasoning graphs.

But decomposition depth must remain controllable.

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## 7. DFS + DAG Decomposition Architecture

### Proposed Solution

Use:

- DFS-style recursive decomposition,
- constrained by DAG graph representation.

This prevents:

- infinite cycles,
  - redundant expansion,
  - semantic duplication.
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### Example

Input:

```
Smoking causes cancer
```

Depth 0:

```
Smoking causes cancer
```



Depth 1:

```
Smoking damages tissue
Smoking correlates with cancer
Smoking precedes cancer
```

Depth 2:

```
DNA mutation
Oxidative stress
Cellular damage
Carcinogens
```

---

## DAG Benefits

DAG representation allows:

- shared reasoning motifs,
- subgraph reuse,
- contradiction propagation,
- graph compression,
- cache efficiency.

Example:

```
Smoking → DNA damage → Cancer
Radiation → DNA damage → Cancer
```

Both reuse:

```
DNA damage
```

instead of duplicating decomposition trees.

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## Required Additions

### Cycle Detection

Needed for:

Economy ↔ Politics

style loops.

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## Semantic Deduplication

Example:

DNA damage  
Genetic damage  
Mutation damage

may represent identical nodes.

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## Information Gain Heuristics

Only expand decomposition if:

- uncertainty reduces,
- contradiction unresolved,
- information gain increases.

This prevents infinite semantic expansion.

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## 8. MultiLensRouter Evolution

### Original Role

Originally:

| Expert selection and routing.

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### New Realization

The MultiLensRouter is evolving into:

| A semantic arbitration system.

It now performs:

- semantic interpretation,
  - structural arbitration,
  - confidence fusion,
  - ontology grounding,
  - object-level extraction,
  - semantic disambiguation.
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## 9. Existing Router Architecture

Current lenses:

### Lens 1 -- Semantic Similarity

Sentence-transformer semantic routing.

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### Lens 2 -- Spectral Structural Analysis

Spectral signature matching using:

- vocabulary structure,
  - syntactic patterns,
  - domain-specific linguistic profiles.
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### Lens 3 -- Confidence Fusion

Combines:

- semantic scores,
  - spectral scores,
  - confidence priors.
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## 10. New Router Responsibilities

The router can evolve into:

- ontology reconciliation,
  - semantic canonicalization,
  - temporal interpretation,
  - decomposition arbitration,
  - contradiction classification,
  - relation-family routing.
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## 11. Relation-Level Routing

### Proposed Upgrade

Route not only domains,  
but also:

```
Causes  
CorrelatesWith  
Contradicts  
Supports  
Predicts  
Precedes  
Implies
```

This enables:

- causal reasoning,
- probabilistic reasoning,
- temporal reasoning,
- ethical reasoning,
- contradiction semantics.

Different relation families obey different inference rules.

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## 12. Epistemic Polarity Signatures

# Proposed Spectral Upgrade

Add spectral axes for:

- certainty,
- speculation,
- manipulation,
- propaganda,
- ideological framing,
- adversarial phrasing,
- probabilistic language.

Example:

This definitively proves...

vs

Evidence weakly suggests...

These have radically different epistemic structures.

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## 13. Object-Level Extraction

Current object-level extraction is effectively:

Proto-ontology reasoning.

Example:

```
String Theory
  ↓
Theoretical Physics
  ↓
Physics
  ↓
Science
```

This supports:

- abstraction layering,
  - decomposition grounding,
  - ontology inheritance,
  - evidence retrieval.
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## 14. TRM Integration

### Core Realization

TRM and MultiLensRouter solve different problems.

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### MultiLensRouter Role

Best at:

- semantic interpretation,
- ambiguity arbitration,
- relation-family inference,
- semantic routing.

Answers:

“What does this probably mean?”

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### TRM Role

Best at:

- longitudinal stabilization,
- semantic persistence,
- canonical graph convergence,
- reasoning memory consolidation,
- decomposition stabilization.

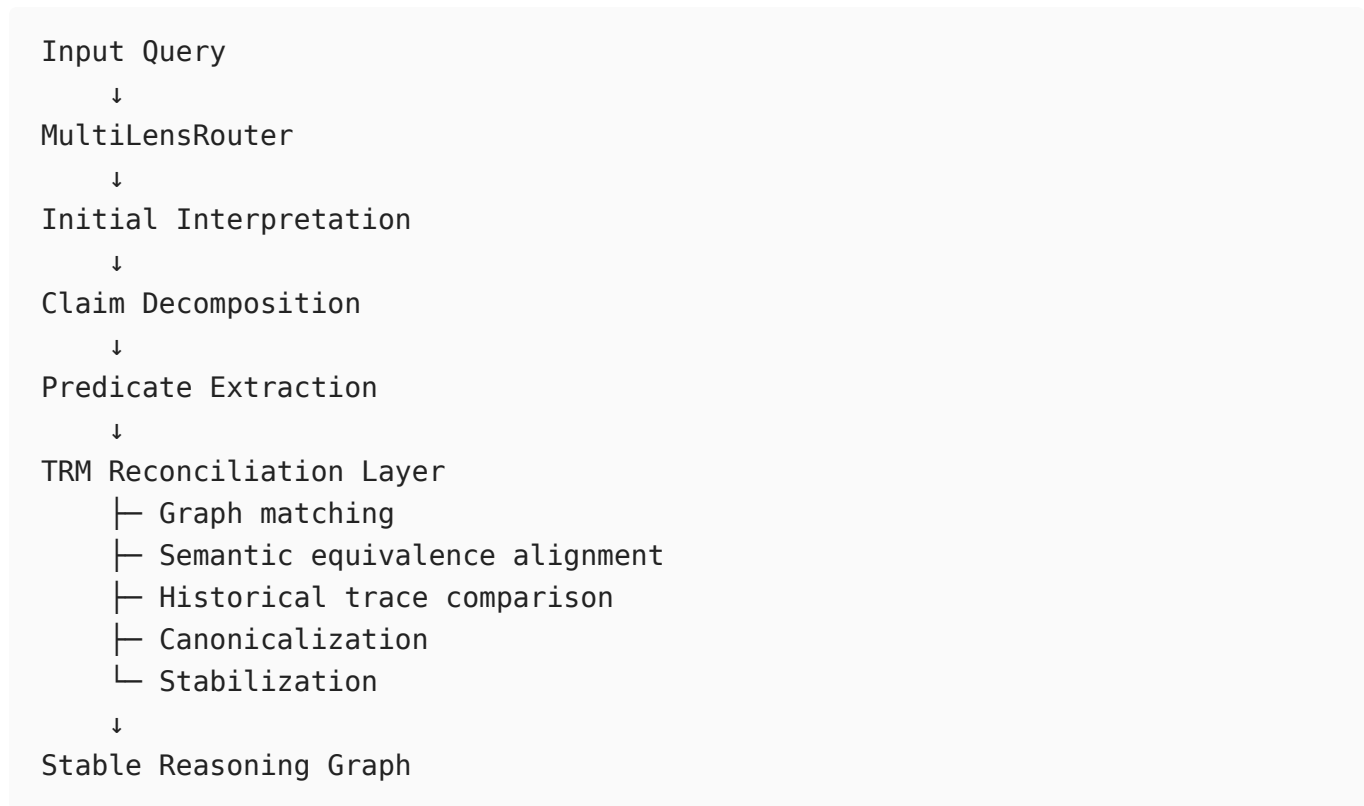
Answers:

“How has this meaning historically stabilized?”

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## 15. TRM as Semantic Convergence Layer

### Proposed Flow



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### Purpose

Without stabilization:

- decomposition drifts,
- ontology entropy grows,
- graph reuse collapses,
- contradiction analysis weakens.

TRM prevents this through semantic convergence.

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## 16. Canonical Graph Fingerprints

### Proposed Addition

Every stabilized reasoning graph/subgraph should generate:

```
graph_fingerprint = {  
  "semantic_hash": "...",  
  "structural_hash": "...",  
  "predicate_family_hash": "...",  
  "temporal_signature": "..."  
}
```

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## Benefits

Supports:

- graph deduplication,
- cache reuse,
- contradiction lineage,
- semantic stabilization,
- decomposition convergence.

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## 17. Decomposition Attractors

TRM can reinforce recurring reasoning structures.

Example:

```
Smoking → DNA damage → Cancer
```

If rediscovered repeatedly,

TRM can consolidate it into:

| A canonical reusable reasoning motif.

This enables:

- graph compression,
- memory consolidation,
- reasoning reuse,
- longitudinal stabilization.



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## 18. Contradiction Semantics

### Remaining Open Problem

Not all disagreement is contradiction.

Example:

```
Coffee improves focus  
Coffee increases anxiety
```

These are NOT contradictory.

Future contradiction ontology should support:

- direct contradiction,
- contextual contradiction,
- probabilistic disagreement,
- conditional contradiction,
- temporal contradiction.

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## 19. Evidence Lineage

All conclusions should remain traceable.

Example:

```
Claim  
→ predicate  
→ evidence  
→ source  
→ decomposition path  
→ reasoning chain
```

Without lineage:

- hallucination propagation becomes opaque,
- trust collapses,
- reasoning becomes non-auditable.

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## 20. Memory Updating

### Future Requirement

When new evidence arrives:

```
Old evidence → X  
New evidence → Y
```

The system must determine:

- invalidate?
- weaken confidence?
- maintain competing hypotheses?
- version reasoning graphs?

This becomes a core cognitive architecture problem.

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## 21. Multi-Expert Conflict Resolution

Future issue:

Different experts may:

- use incompatible ontologies,
- disagree on evidence standards,
- reason at different abstraction levels.

The system will eventually require:

| Meta-reasoning and ontology arbitration.

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## 22. Determinism Requirement

As the system becomes:

- dynamic,
- semantic,

- adaptive,
- probabilistic,

stable determinism becomes increasingly important.

Needed for:

- cache consistency,
- reproducibility,
- debugging,
- longitudinal graph comparison,
- contradiction tracing.

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## **23. Recommended Future Stabilization Mechanisms**

### **Canonicalization Passes**

Normalize graphs before final storage.

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### **Deterministic Graph Ordering**

Ensure stable serialization.

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### **Stable Decomposition Policies**

Prevent decomposition chaos.

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### **Canonical Semantic Equivalence Classes**

Reduce ontology fragmentation.

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## **24. Final Architectural Realization**

Mycelium is gradually evolving toward:

- semantic memory consolidation,
- ontology stabilization,
- epistemic arbitration,
- graph-based reasoning,
- uncertainty-aware cognition,
- longitudinal semantic persistence.

The combined system:

- MultiLensRouter
- TRM
- DAG decomposition
- contradiction reasoning
- evidence grounding
- reasoning cache
- ontology stabilization

collectively forms:

| A semantic operating system for reasoning.